

Multiplicity Moonshine: Prime-Resolved Fixed-Point Dynamics and Modular Invariants

Ryan O. van Gelder & Tyler van Osdol

Citizen Gardens

The Foundation of Multiplicity

April 4, 2026

Abstract

This report synthesizes a series of developments linking Multiplicity Theory—a prime-indexed, recursively stable operator framework—with Monstrous Moonshine and modular forms. We formulate a precise “Multiplicity Moonshine” principle: modular q -series emerge as invariants of a prime-graded fixed-point dynamical system driven by Hecke-type operators under strict projector, conservation, and contraction constraints. We define the associated computational validation engine, including (i) a prime-word and $\mathbb{Z}_{\geq 0}^P$ -graded state space, (ii) an ACE+projector+PETC iteration admitting a unique fixed point, (iii) a Hecke algebra layer over this space, and (iv) a modularity test suite based on S and Fricke transforms. We present a strong conjectural bridge between the prime-resolved multiplicity field and classical Moonshine coefficients, and specify a production-grade protocol for falsifying or confirming this bridge. Code-level sketches for a validation package are included.

Contents

1	Executive Summary	4
2	Prime-Graded State Space and Projectors	5
2.1	Prime-Graded Hilbert Space	5
2.2	Prime Signatures and $\mathbb{Z}^P$ -Grading	5
3	ACE+Projector+PETC Runtime and Fixed Point	6
3.1	Channels, Gains, ACE Constraint	6
3.2	Block Commutation and Invariants	6
3.3	Projected Fixed-Point Iteration	6
4	Prime-Word Representation and Ω-Projection	7
4.1	Prime-Word Representation of the Fixed Point	7
4.2	Compressed Ω -Graded Coefficients	7
5	Hecke Algebra on the Prime-Graded Space	7
5.1	Hecke Operators on Truncated q -Series	8
5.2	Hecke Multiplicativity and Eigenfunctions	8
6	Multiplicity Moonshine Conjecture	8
7	Modularity Test Suite	9
7.1	Analytic Tests: S and T	9
7.2	Level and Fricke Involutions	9
7.3	External Matching	9
8	Runtime Diagnostics as Order Parameters	10
9	Computational Validation Engine	10
9.1	Prime-Word and Ω -Grading Layer	10
9.2	Fixed-Point Iteration with Rational Certification	11
9.3	Hecke Algebra Layer	11
9.4	Modularity Tests	12
10	Recommended Extensions and Further Work	12
10.1	Automated Reporting	12
10.2	Search over Configurations	12
10.3	Symmetry Group Analysis	12
11	Conclusion	13
	Mathematical Appendix	13
A	Operator-Theoretic Setup and Notation	13
A.1	Hilbert Space and Projectors	13
A.2	Channels and ACE Budget	14
A.3	External Term and Certified Window	14
B	Projected Fixed-Point Iteration: Contraction Proof	14
B.1	Single-Step Contraction	15
B.2	Global Contraction and Fixed Point	16
C	Commutation Residual and Block Structure	16
C.1	Block Commutation and Residual	17
C.2	Implication for Diagnostic $\Delta_{p,t}$	17
D	Sector Shares, Manifold Energy, and Entropy	18
D.1	Sector Shares	18
D.2	Manifold Energy	18
D.3	Sector Entropy	19

E	Hecke Operators on Truncated q-Series	19
E.1	Hecke Operator Definition	19
E.2	Operator Norm Bound (Euclidean Norm)	20
E.3	Exact Commutativity and Multiplicativity on $\mathcal{V}_N$	20
F	Summary of Bounds and Conditions	21

1 Executive Summary

The core claim of this work is that Multiplicity Theory, as developed in recent Multiplicity Foundation / Citizen Gardens preprints, already contains the structural skeleton of a generalized Moonshine theory:

- Classical Moonshine uses a $\mathbb{Z}$ -graded vector space $V = \bigoplus_{n=-1}^{\infty} V_n$ equipped with a Monster action, whose graded traces produce modular functions (e.g. the j -function).
- Multiplicity Theory uses a prime-indexed, projector-structured Hilbert space

$$\mathcal{H} = \bigoplus_{p \in P} \mathcal{H}_p$$

together with a runtime-safe evolution

$$T_{t+1} = \Pi(F_t + K_t T_t)$$

in which each K_t is a prime-weighted operator constrained by ACE budgets and PETC conservation, and Π is a lawfulness projector enforcing on-manifold evolution.

The novelty is not a vague analogy but a precise refactor:

- The $\mathbb{Z}$ -grading of classical Moonshine is replaced by a $\mathbb{Z}_{\geq 0}^P$ -grading over prime exponent vectors (“prime words”).
- The integer index n underlying the q -expansion is interpreted as a compressed observable $\Omega(W)$ of an underlying prime-word W , with Ω typically taken as the total prime exponent count.
- Moonshine coefficients become emergent *resonance weights* computed from a fixed-point of the multiplicity evolution.

We formulate a *Multiplicity Moonshine* conjecture: under explicit operator, projector, and Hecke-algebra conditions, the q -series obtained by Ω -compressing the prime-word fixed point is a modular form on a congruence subgroup, and in special cases matches known Moonshine series.

Operationally, we specify a validation engine:

1. Construct a prime-word graded state space and a projector family $\{\Pi_p\}$.
2. Run a projected fixed-point iteration with ACE budget and PETC bookkeeping.
3. Lift the converged float fixed point to exact rationals and certify the residual.
4. Build exact Hecke matrices T_m , verify multiplicativity and simultaneous eigenfunction behavior on the fixed point.
5. Compress to a q -series and run a modularity suite (S-transform, weight fit, Fricke, comparison with known forms).
6. Correlate modularity with runtime diagnostics $(\Delta_{p,t}, \text{ME}(t), H[q_t])$ as an “order parameter” for emergent modular symmetry.

This document consolidates the mathematics and the computational design of such a validator.

2 Prime-Graded State Space and Projectors

We start from the Multiplicity Core blueprint, where the state space and projector structure are defined.

2.1 Prime-Graded Hilbert Space

Let P be a finite set of primes. We work in a (real or complex) Hilbert space $\mathcal{H}$ with norm $\|\cdot\|$ and inner product $\langle \cdot, \cdot \rangle$. We assume a prime-sector decomposition:

$$\mathcal{H} = \bigoplus_{p \in P} \mathcal{H}_p, \quad (1)$$

where each $\mathcal{H}_p$ is a closed subspace corresponding to the “ p -sector.”

Definition 2.1 (Prime Projectors). For each $p \in P$ we assume a bounded linear projector

$$\Pi_p : \mathcal{H} \rightarrow \mathcal{H}$$

satisfying, on a “certified window” of states,

$$\Pi_p^2 = \Pi_p, \quad \Pi_p \Pi_q = 0 \quad (p \neq q), \quad \sum_{p \in P} \Pi_p = \Pi_P, \quad (2)$$

where Π_P projects onto the prime sector span. We also assume nonexpansiveness $\|\Pi_p\| \leq 1$, $\|\Pi_P\| \leq 1$.

An optional scalar projector Π_S can project onto a certified subspace (e.g. a lawfulness manifold). We write $\Pi \in \{\Pi_P, \Pi_S\}$ for the projector used in the core iteration.

2.2 Prime Signatures and $\mathbb{Z}^P$ -Grading

Multiplicity Theory associates a *prime signature* vector $e \in \mathbb{Z}^P$ to typed operators and states. Composition and aggregation are additive in this signature; at runtime PETC enforces that signatures and sector totals update additively.

In the Moonshine setting we refine this into a full-grading picture.

Definition 2.2 (Prime Words and Exponent Vectors). Fix an ordering of $P = \{p_1, \dots, p_r\}$. A *prime word* is an exponent vector

$$e = (k_1, \dots, k_r) \in \mathbb{Z}_{\geq 0}^r.$$

We identify the associated integer

$$W(e) := \prod_{i=1}^r p_i^{k_i}.$$

Define the *prime-word space*

$$\mathbb{Z}_{\geq 0}^P \cong \mathbb{Z}_{\geq 0}^r.$$

Definition 2.3 (Ω -Grading). Define the Ω -value of a prime word by

$$\Omega(e) := \sum_{i=1}^r k_i. \quad (3)$$

We obtain an induced grading

$$\mathcal{H} = \bigoplus_{n \geq 0} \mathcal{H}_n^{(\Omega)}, \quad \mathcal{H}_n^{(\Omega)} := \bigoplus_{\Omega(e)=n} \mathcal{H}_e, \quad (4)$$

where $\mathcal{H}_e$ denotes the subspace corresponding to exponent vector e (or, in practice, the subspace spanned by basis vectors indexed by e).

In this way, the $\mathbb{Z}$ -grading familiar from Moonshine is a *compression* of a richer $\mathbb{Z}_{\geq 0}^P$ -grading.

3 ACE+Projector+PETC Runtime and Fixed Point

We now recall the runtime iteration and its convergence guarantee.

3.1 Channels, Gains, ACE Constraint

Let $\{B_p\}_{p \in P} \subset L(\mathcal{H})$ be a family of bounded linear operators (channels) with known bounds $\|B_p\| \leq b_p$. For each step t , a PETC-derived weight vector $w_{p,t} \in \mathbb{R}$ defines the gain operator

$$K_t = \sum_{p \in P} w_{p,t} B_p. \quad (5)$$

The ACE budget imposes

$$\sum_{p \in P} b_p |w_{p,t}| \leq \tau_t, \quad 0 \leq \tau_t < 1, \quad (6)$$

with a global cap $\tau_t \leq \tau_{\max} < 1$ on the certified window.

An external term $F_t : \mathcal{H} \rightarrow \mathcal{H}$ is assumed bounded and Lipschitz on the certified window (often instantiated as a Hecke/replicability mixer).

3.2 Block Commutation and Invariants

For each audited update operator U_t used to build F_t or K_t , we enforce block commutation with prime projectors:

$$U_t \Pi_p = \Pi_p U_t \quad \text{for all } p \in P \quad (7)$$

on the certified window, up to a controllable residual

$$\Delta_{p,t} := \|U_t \Pi_p - \Pi_p U_t\|, \quad \max_{p \in P} \Delta_{p,t} \leq \eta. \quad (8)$$

3.3 Projected Fixed-Point Iteration

Let $T_t \in \mathcal{H}$ denote the state at time t . The core iteration is

$$T_{t+1} = \Pi(F_t + K_t T_t). \quad (9)$$

Define sector shares

$$q_{p,t} = \frac{\|\Pi_p T_t\|^2}{\sum_{r \in P} \|\Pi_r T_t\|^2}, \quad \sum_{p \in P} q_{p,t} = 1, \quad (10)$$

with the convention $0/0 := 0$. Define manifold energy

$$\text{ME}(t) = \frac{\|\Pi_P T_t\|^2}{\|T_t\|^2 + \varepsilon} \quad (11)$$

and sector entropy

$$H[q_t] = - \sum_{p \in P} q_{p,t} \log q_{p,t}. \quad (12)$$

Acceptance gates enforce

$$\tau_t \leq \tau_{\max} < 1, \quad \text{ME}(t) \geq 1 - \epsilon_{\text{ME}}, \quad H[q_t] \in [H_{\min}, H_{\max}], \quad \max_p \Delta_{p,t} \leq \eta. \quad (13)$$

Theorem 3.1 (Stepwise Contraction and Fixed Point). *Assume that the ACE budget holds with margin $1 - \tau_t > 0$, Π is nonexpansive, F_t is Lipschitz and bounded on the certified window, and all operators in the step commute with $\{\Pi_p\}$ up to bounded residuals as above. Let $\widehat{K}_t$ denote the realized linear part at step t , and suppose $\|\widehat{K}_t\| \leq \tau_t < 1$ for all t . Then, restricted to the certified window, the iteration (9) defines a contraction mapping admitting a unique fixed point T^* , and for any two trajectories $T_t, \tilde{T}_t$ driven by the same schedule,*

$$\|T_{t+1} - \tilde{T}_{t+1}\| \leq \tau_t \|T_t - \tilde{T}_t\|.$$

This provides the rigorous foundation for using T^* as the multiplicity field from which Moonshine coefficients will be derived.

4 Prime-Word Representation and Ω -Projection

We now make the prime-word structure explicit.

4.1 Prime-Word Representation of the Fixed Point

In many relevant Moonshine workloads (e.g. q -series with indices $n \geq -1$), one works with a truncated basis indexed by integers. Under the prime-word interpretation, we may equivalently represent the fixed point T^* as a sparse map

$$T^* : \mathbb{Z}_{\geq 0}^P \rightarrow \mathbb{Q}$$

sending a prime word e to a rational coefficient $T^*(e)$. In practice this is stored as a dictionary

$$\{e \mapsto T^*(e)\}_{e: \Omega(e) \leq \Omega_{\max}}$$

for some truncation bound.

4.2 Compressed Ω -Graded Coefficients

Definition 4.1 (Ω -Compressed Coefficients). Given $T^* : \mathbb{Z}_{\geq 0}^P \rightarrow \mathbb{C}$, define the Ω -graded coefficients by

$$c_\Omega(n) := \sum_{\Omega(e)=n} |T^*(e)|^2 \tag{14}$$

for $n \geq 0$. We define the associated q -series

$$\chi_{\mathcal{M}}(\tau) := \sum_{n \geq n_0} c_\Omega(n) q^n, \quad q = e^{2\pi i \tau}, \tag{15}$$

for an appropriate starting index n_0 (e.g. $n_0 = -1$ for j -like series).

This construction realizes the idea that classical Moonshine coefficients are a “shadow” of a higher-resolution prime-resolved state: each $c_\Omega(n)$ aggregates all prime words with total exponent sum n .

5 Hecke Algebra on the Prime-Graded Space

We now specify the Hecke layer that connects to modular forms.

5.1 Hecke Operators on Truncated q -Series

For concreteness, consider the truncated space of q -series coefficients $a(n)$ for $0 \leq n \leq N$. The classical Hecke operator T_p acts (in an appropriate normalization) by

$$(T_p a)[n] = a[pn] + \mathbf{1}_{p|n} a[n/p], \quad (16)$$

with suitable truncation boundary conditions at $n > N$. This can be represented as an integer-valued matrix $T_p \in \mathbb{Z}^{(N+1) \times (N+1)}$.

For composite m with factorization $m = \prod p^{\ell_p}$, we build T_m from the prime T_p by composition and Hecke algebra relations. On the truncated space, we require these relations to hold exactly as integer matrix identities.

5.2 Hecke Multiplicativity and Eigenfunctions

Definition 5.1 (Hecke Algebra Conditions). Fix a truncation bound M on the Hecke index and a bound N on the q -series. We require:

(H1) For all coprime $m, n \leq M$,

$$T_m T_n = T_n T_m, \quad T_m T_n = T_{mn}$$

as matrix equalities on the truncated space.

(H2) The fixed point T^* , mapped to the coefficient vector $(a(0), \dots, a(N))$, is a simultaneous Hecke eigenvector for all $m \leq M$:

$$T_m T^* = \lambda_m T^*, \quad \lambda_m \in \mathbb{Q}. \quad (17)$$

These are the algebraic conditions expected of modular forms with respect to the Hecke algebra.

6 Multiplicity Moonshine Conjecture

We can now state a clean conjecture capturing the bridge from Multiplicity dynamics to modular forms.

Conjecture 6.1 (Multiplicity Moonshine Principle (Strong Form)). *Let P be a finite set of primes and let $\mathcal{H}$ be a prime-graded Hilbert space with exact orthogonal projectors $\{\Pi_p\}_{p \in P}$ and lawfulness projector Π as above. Let T^* be the unique fixed point of the ACE+projector+PETC iteration*

$$T_{t+1} = \Pi(F + K T_t)$$

on the certified window, with F a Hecke-style arithmetic mixer and K an ACE-constrained prime-weighted operator.

Assume:

(C1) (Exact projectors) $\Pi_p \Pi_q = 0$ for $p \neq q$, $\sum_p \Pi_p = \Pi_P$.

(C2) (Hecke eigencondition) *The q -series coefficients derived from T^* form a simultaneous Hecke eigenfunction for all T_m up to some M :*

$$T_m T^* = \lambda_m T^*, \quad \lambda_m \in \mathbb{Q}.$$

(C3) (Hecke multiplicativity) *The operators satisfy $T_m T_n = T_{mn}$ and commute for all coprime $m, n \leq M$.*

(C4) (Ω -grading) The q -series $\chi_{\mathcal{M}}(\tau) = \sum_{n \geq n_0} c_{\Omega}(n)q^n$ is obtained by Ω -compression of the prime-word representation of T^* .

Then there exist level N and weight k such that

$$\chi_{\mathcal{M}}(\tau) \in M_k(\Gamma_0(N)),$$

i.e. $\chi_{\mathcal{M}}$ is a modular form (or a weakly holomorphic modular function) for a congruence subgroup.

In special configurations of (P, F, K, Π) , one expects the coefficients $c_{\Omega}(n)$ to match known Moonshine series (e.g. the j -function or McKay–Thompson series) up to normalization.

7 Modularity Test Suite

In computational practice, modularity is tested via transforms on the upper half-plane and comparison with known databases.

7.1 Analytic Tests: S and T

Let $\chi(\tau) = \sum_{n=n_0}^N c(n)q^n$ denote the truncated candidate series.

The modular group $\mathrm{SL}_2(\mathbb{Z})$ is generated by

$$S : \tau \mapsto -1/\tau, \quad T : \tau \mapsto \tau + 1.$$

T -invariance is automatic for integer exponents. For S -invariance of a weight k modular form, we expect

$$\chi(-1/\tau) = \tau^k \chi(\tau). \tag{18}$$

Numerically, one evaluates χ on several τ in the upper half-plane and estimates the best-fit k by minimizing

$$E_k(\tau) := |\chi(-1/\tau) - \tau^k \chi(\tau)|$$

over candidate weights.

7.2 Level and Fricke Involutions

For a congruence subgroup $\Gamma_0(N)$, one can test eigenbehavior under Atkin–Lehner involutions, particularly the Fricke involution $W_N : \tau \mapsto -1/(N\tau)$. A modular form of weight k and Fricke eigenvalue ϵ_{W_N} satisfies

$$\chi(-1/(N\tau)) = \epsilon_{W_N} (N\tau)^k \chi(\tau).$$

Practically, one estimates ϵ_{W_N} and checks the magnitude of the residual across sample points.

7.3 External Matching

If the analytic tests suggest modularity, one can compare the first several coefficients $c(n)$ against known modular forms catalogued in databases such as the LMFDB, or via symbolic tools (Sage, PARI) to see if χ lies in the span of known basis forms.

8 Runtime Diagnostics as Order Parameters

An important empirical prediction is that certain runtime diagnostics of the Multiplicity iteration serve as *order parameters* for emergent modular symmetry.

Define:

- Commutation residual:

$$\Delta_t := \max_{p \in P} \|U_t \Pi_p - \Pi_p U_t\|,$$

where U_t is the audited update operator.

- Manifold energy:

$$\text{ME}(t) := \frac{\|\Pi_P T_t\|^2}{\|T_t\|^2 + \varepsilon}.$$

- Sector entropy:

$$H[q_t] := - \sum_{p \in P} q_{p,t} \log q_{p,t}.$$

Conjecture 8.1 (Diagnostic Modularity Criterion). *For Moonshine-targeted workloads and fixed (P, N, F, K, Π) , the candidate q -series $\chi_{\mathcal{M}}$ converges to a modular form if and only if, along the ACE+projector+PETC iteration,*

$$\lim_{t \rightarrow \infty} \Delta_t = 0, \quad \lim_{t \rightarrow \infty} \text{ME}(t) = 1, \quad H[q_t] \rightarrow H_{\infty}$$

for some finite entropy H_{∞} .

This connects algebraic modularity to purely runtime-observable quantities.

9 Computational Validation Engine

We now describe a minimal, code-level architecture for a validation engine implementing the above mathematics. Language and libraries are left open; the structure is what matters.

9.1 Prime-Word and Ω -Grading Layer

Data Structures

- Represent a prime word as a tuple of exponents

$$e = (k_1, \dots, k_r) \in \mathbb{Z}_{\geq 0}^r.$$

- Store the fixed point T^* as a sparse map

$$\{e \mapsto T^*(e)\}.$$

- Maintain a grouped representation

$$\{n \mapsto \{e : \Omega(e) = n \mapsto T^*(e)\}\}.$$

Key Operations

- $\Omega(e)$: total exponent sum.
- $W(e)$: integer value of the word.
- Projection to Ω :

$$c_{\Omega}(n) = \sum_{\Omega(e)=n} |T^*(e)|^2.$$

9.2 Fixed-Point Iteration with Rational Certification

Float Iteration

Implement the ACE+projector+PETC iteration in floating arithmetic:

```
Initialize T = T0
for t in 0..max_iter:
    w_t = PETC.propose_weights(T)
    enforce ACE budget on w_t
    K_t = sum_p w_t * B_p
    U_t, F_t = build_update(T)
    enforce commutation residual bound on U_t
    T_next = Pi(F_t + K_t * T)
    compute diagnostics (Delta_t, ME(t), H[q_t])
    if ||T_next - T|| < tol:
        break
    T = T_next
```

Rational Lift

After convergence of the float iteration:

- Map the float coordinates of T to rational numbers (via rational approximation or a rational backend if available from the start).

- Compute

$$R = T_{\mathbb{Q}}^* - \Pi(F + KT_{\mathbb{Q}}^*)$$

exactly, and derive a rational or norm-based residual.

- Declare the fixed point certified if R is exactly zero or has norm below a predetermined threshold with rigorous control.

9.3 Hecke Algebra Layer

Matrix Construction

For a truncation bound N and a prime set P :

- Construct exact integer matrices T_p for each $p \in P$ on indices $0 \leq n \leq N$ according to the chosen Hecke stencil.
- For a given m , construct T_m via prime factorization and composition (respecting Hecke relations).

Algebraic Tests

- For all coprime $m, n \leq M$ in the test set, verify exact equality

$$T_m T_n = T_n T_m, \quad T_m T_n = T_{mn}.$$

- For the fixed-point coefficient vector T^* , solve

$$T_m T^* = \lambda_m T^*$$

for λ_m using exact linear algebra; check that $\lambda_m \in \mathbb{Q}$.

9.4 Modularity Tests

Given the compressed q -series coefficients $c(n)$:

- Implement a function to evaluate $\chi(\tau)$ at arbitrary τ with high precision.
- Evaluate S-transform residuals $E_k(\tau)$ over sample points and candidate weights.
- If small residuals are found, estimate the best-fit weight k .
- Optionally, evaluate Fricke/Atkin–Lehner residuals to infer level N .
- Dump the first several coefficients $c(n)$ for external comparison against modular form databases.

10 Recommended Extensions and Further Work

10.1 Automated Reporting

Beyond the core validation pipeline, one may add:

- Automatically generated plots of diagnostic trajectories $(\Delta_t, \text{ME}(t), H[q_t])$.
- Tabular comparison of $c_\Omega(n)$ with known Moonshine coefficients (e.g. those of the j -function) for early n .
- Automated generation of a PDF report summarizing configuration, results, and evidence.

10.2 Search over Configurations

One can scan over:

- Different prime sets P (e.g. $\{2, 3\}$, $\{2, 3, 5\}$).
- Different truncation levels N and $\Omega_{\max}$.
- Different choices of F (Hecke patterns, replicability maps).
- Different projectors Π_S defining lawfulness manifolds.

to map out where modular behavior does or does not emerge.

10.3 Symmetry Group Analysis

Defining the automorphism group

$$G_{\mathcal{M}} = \{U \in L(\mathcal{H}) : U\Pi_p = \Pi_p U \ \forall p, \ [U, T_m] = 0 \ \forall m\}$$

and studying its finite subgroups may reveal sporadic structures. In particular, one may search computationally for large finite simple subgroups consistent with known sporadic orders.

11 Conclusion

We have presented a mathematically coherent and computationally precise bridge between Multiplicity Theory and Moonshine. The key idea is that modular q -series can be realized as Ω -compressed invariants of a prime-graded fixed-point system governed by strict projector, ACE, and conservation constraints and endowed with a Hecke algebra action.

The resulting *Multiplicity Moonshine* principle is falsifiable: given a finite prime set, truncation bounds, a concrete arithmetic mixer F , and runtime parameters, the validation engine either produces a certified modular series (and possibly a known Moonshine match), or it does not. The same machinery also yields a rich set of runtime diagnostics and symmetry candidates. Taken together, these developments suggest that classical Monstrous Moonshine may indeed be a low-resolution shadow of a higher-resolution multiplicative structure.

Mathematical Appendix

A Operator-Theoretic Setup and Notation

We collect the functional-analytic assumptions and notational conventions used throughout the main text and give explicit operator norm bounds for the projected fixed-point iteration

$$T_{t+1} = \Pi(F_t + K_t T_t).$$

A.1 Hilbert Space and Projectors

Let $\mathcal{H}$ be a complex Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\|\cdot\|$. We write $L(\mathcal{H})$ for the bounded linear operators on $\mathcal{H}$, equipped with the operator norm

$$\|A\|_{\text{op}} := \sup_{\|x\|=1} \|Ax\|.$$

Definition A.1 (Prime Sector Projectors). Let P be a finite set of primes. A *prime sector projector family* on $\mathcal{H}$ consists of operators

$$\Pi_p \in L(\mathcal{H}), \quad p \in P,$$

and a combined prime projector $\Pi_P \in L(\mathcal{H})$ such that:

1. $\Pi_p^2 = \Pi_p$ (idempotence);
2. $\Pi_p \Pi_q = 0$ for $p \neq q$ (orthogonality);
3. $\sum_{p \in P} \Pi_p = \Pi_P$ (partition of the prime sector);
4. $\|\Pi_p\|_{\text{op}} \leq 1$ and $\|\Pi_P\|_{\text{op}} \leq 1$.

The projector Π used in the iteration is either Π_P or an additional “lawfulness” projector Π_S (with $\|\Pi_S\|_{\text{op}} \leq 1$), but for the purposes of the proofs below we only require that Π be nonexpansive:

$$\|\Pi\|_{\text{op}} \leq 1.$$

A.2 Channels and ACE Budget

We model channel operators and their combination as follows.

Definition A.2 (Channel Family and ACE Budget). Let $\{B_p\}_{p \in P} \subset L(\mathcal{H})$ be a family of bounded linear operators, with known bounds

$$\|B_p\|_{\text{op}} \leq b_p, \quad b_p \geq 0.$$

For each time step t , let $w_{p,t} \in \mathbb{R}$ be scalar weights and define

$$K_t := \sum_{p \in P} w_{p,t} B_p.$$

We say that the *ACE budget* is satisfied at time t with margin $1 - \tau_t > 0$ if

$$\sum_{p \in P} b_p |w_{p,t}| \leq \tau_t < 1.$$

By submultiplicativity and the triangle inequality we obtain a uniform bound on $\|K_t\|_{\text{op}}$ in terms of τ_t .

Proposition A.3 (Operator Norm Bound for K_t). *Under the ACE budget assumption,*

$$\|K_t\|_{\text{op}} = \left\| \sum_{p \in P} w_{p,t} B_p \right\|_{\text{op}} \leq \sum_{p \in P} |w_{p,t}| \|B_p\|_{\text{op}} \leq \tau_t < 1.$$

Proof. For any $x \in \mathcal{H}$ with $\|x\| = 1$,

$$\|K_t x\| = \left\| \sum_{p \in P} w_{p,t} B_p x \right\| \leq \sum_{p \in P} |w_{p,t}| \|B_p x\| \leq \sum_{p \in P} |w_{p,t}| b_p.$$

Taking the supremum over all $\|x\| = 1$ and using the ACE budget inequality completes the proof. $\square$

A.3 External Term and Certified Window

We allow for a time-dependent external term $F_t \in \mathcal{H}$ or, more generally, an affine map $F_t : \mathcal{H} \rightarrow \mathcal{H}$ that is Lipschitz on a prescribed subset of $\mathcal{H}$.

Definition A.4 (Certified Window and Lipschitz External Term). Let $W \subset \mathcal{H}$ be a closed, convex subset (the *certified window*) such that all iterates T_t lie in W . We assume:

- $F_t : W \rightarrow \mathcal{H}$ is L_F -Lipschitz for each t , i.e.,

$$\|F_t(x) - F_t(y)\| \leq L_F \|x - y\| \quad \forall x, y \in W;$$

- F_t is uniformly bounded on W : $\sup_{x \in W} \|F_t(x)\| \leq M_F < \infty$.

In many of the workloads of interest, F_t is independent of t or depends only mildly on t ; the proofs below make no such restriction beyond uniform Lipschitzness and boundedness on W .

B Projected Fixed-Point Iteration: Contraction Proof

We now make precise and prove the contraction property of the map

$$\Phi_t(T) := \Pi(F_t + K_t T),$$

and deduce the existence and uniqueness of a fixed point under a uniform contraction bound.

B.1 Single-Step Contraction

Proposition B.1 (Contraction of Φ_t on the Certified Window). *Assume:*

1. $\Pi \in L(\mathcal{H})$ is nonexpansive, $\|\Pi\|_{\text{op}} \leq 1$.
2. $K_t \in L(\mathcal{H})$ satisfies $\|K_t\|_{\text{op}} \leq \tau_t < 1$.
3. F_t is fixed (does not depend on T) or, more generally, satisfies $F_t(x) - F_t(y) = 0$ whenever $x, y \in W$ (e.g. is constant on the window).

Then for all $T_1, T_2 \in W$,

$$\|\Phi_t(T_1) - \Phi_t(T_2)\| \leq \tau_t \|T_1 - T_2\|.$$

Proof. We compute:

$$\begin{aligned} \|\Phi_t(T_1) - \Phi_t(T_2)\| &= \|\Pi(F_t + K_t T_1) - \Pi(F_t + K_t T_2)\| \\ &= \|\Pi(K_t T_1 - K_t T_2)\| \quad (\text{since the } F_t \text{ terms cancel}) \\ &\leq \|\Pi\|_{\text{op}} \|K_t T_1 - K_t T_2\| \\ &\leq \|\Pi\|_{\text{op}} \|K_t\|_{\text{op}} \|T_1 - T_2\| \\ &\leq \tau_t \|T_1 - T_2\|, \end{aligned}$$

using the nonexpansiveness of Π and the bound $\|K_t\|_{\text{op}} \leq \tau_t$. □

In many realistic settings F_t is not strictly constant in T . We then treat the more general affine map

$$S_t(T) := F_t(T) + K_t T$$

and obtain a contraction bound that includes the Lipschitz constant of F_t .

Proposition B.2 (Contraction with Lipschitz External Term). *Assume the conditions above, except now F_t is L_F -Lipschitz on W . Define*

$$\Phi_t(T) := \Pi(F_t(T) + K_t T).$$

Then for all $T_1, T_2 \in W$,

$$\|\Phi_t(T_1) - \Phi_t(T_2)\| \leq (L_F + \tau_t) \|T_1 - T_2\|.$$

In particular, if $L_F + \tau_t < 1$, then Φ_t is a contraction on W .

Proof. We have

$$\begin{aligned} \|\Phi_t(T_1) - \Phi_t(T_2)\| &= \|\Pi(F_t(T_1) + K_t T_1) - \Pi(F_t(T_2) + K_t T_2)\| \\ &\leq \|\Pi\|_{\text{op}} \|F_t(T_1) - F_t(T_2) + K_t(T_1 - T_2)\| \\ &\leq \|F_t(T_1) - F_t(T_2)\| + \|K_t\|_{\text{op}} \|T_1 - T_2\| \\ &\leq L_F \|T_1 - T_2\| + \tau_t \|T_1 - T_2\| \\ &= (L_F + \tau_t) \|T_1 - T_2\|, \end{aligned}$$

using $\|\Pi\|_{\text{op}} \leq 1$ and $\|K_t\|_{\text{op}} \leq \tau_t$. □

B.2 Global Contraction and Fixed Point

We now consider the time-homogeneous case for conceptual clarity: $F_t = F$, $K_t = K$, and $\Phi_t = \Phi$ independent of t . The ACE budget condition reduces to $\|K\|_{\text{op}} \leq \tau < 1$, and we assume F is either constant or Lipschitz with constant L_F .

Theorem B.3 (Banach Fixed-Point Theorem for the Projected Iteration). *Let $W \subset \mathcal{H}$ be nonempty, closed, and convex, and let $\Phi : W \rightarrow W$ be defined by*

$$\Phi(T) := \Pi(F(T) + KT)$$

with Π nonexpansive, $\|K\|_{\text{op}} \leq \tau < 1$, and F either constant or Lipschitz with constant $L_F \geq 0$ such that

$$L_F + \tau < 1.$$

Then:

1. Φ is a contraction on W with contraction factor $\rho := L_F + \tau < 1$.
2. There exists a unique fixed point $T^* \in W$ satisfying

$$T^* = \Pi(F(T^*) + KT^*).$$

3. For any initial $T_0 \in W$, the iterates defined by $T_{t+1} = \Phi(T_t)$ converge to T^* with geometric rate:

$$\|T_t - T^*\| \leq \rho^t \|T_0 - T^*\|, \quad t = 0, 1, 2, \dots$$

Proof. The contraction estimate follows from the previous proposition. The Banach fixed-point theorem (also known as the contraction mapping principle) then guarantees existence and uniqueness of the fixed point and geometric convergence of the iterates. $\square$

Remark B.4 (Extension to Time-Dependent Schedule). If F_t and K_t depend on t but satisfy uniform bounds

$$L_F + \sup_t \|K_t\|_{\text{op}} \leq \rho < 1,$$

and each $\Phi_t(T) := \Pi(F_t(T) + K_t T)$ maps W into itself, then the composition

$$T_{t+1} = \Phi_t(T_t), \quad t = 0, 1, 2, \dots,$$

defines a Cauchy sequence for any initial $T_0 \in W$, and all trajectories collapse into a unique limit point that is independent of the initialization but may depend on the schedule. Specific sufficient conditions are standard in the theory of non-autonomous dynamical systems; we omit the details here since the Moonshine workloads of interest are typically time-homogeneous or periodic.

C Commutation Residual and Block Structure

We now make precise the notion of block commutation with prime projectors and provide explicit bounds on the commutation residual.

C.1 Block Commutation and Residual

Definition C.1 (Block Commutation Residual). Let $U \in L(\mathcal{H})$ be an operator and let Π_p be the prime sector projectors. We define the commutation residual at prime p by

$$\Delta_p(U) := \|U\Pi_p - \Pi_p U\|_{\text{op}}.$$

The maximal commutation residual is

$$\Delta(U) := \max_{p \in P} \Delta_p(U).$$

The idealized “block-diagonal” situation corresponds to $\Delta(U) = 0$. In practice, small but nonzero residuals arise; the following proposition shows that if $\Delta(U)$ is small, the sector-wise action of U remains well-controlled.

Proposition C.2 (Stability of Sectorwise Action Under Small Residuals). *Let $U \in L(\mathcal{H})$ and fix $p \in P$. Then for any $x \in \mathcal{H}$,*

$$\|\Pi_p U x - U \Pi_p x\| \leq \Delta_p(U) \|x\|.$$

Moreover, if $\|U\|_{\text{op}} \leq B$, then

$$\|\Pi_p U \Pi_p x\| \leq (B + \Delta_p(U)) \|\Pi_p x\|.$$

Proof. The first inequality is immediate:

$$\|\Pi_p U x - U \Pi_p x\| = \|(U \Pi_p - \Pi_p U)x\| \leq \|U \Pi_p - \Pi_p U\|_{\text{op}} \|x\| = \Delta_p(U) \|x\|.$$

For the second statement, note that

$$\Pi_p U \Pi_p x = U \Pi_p x + (\Pi_p U \Pi_p x - U \Pi_p x).$$

Thus

$$\begin{aligned} \|\Pi_p U \Pi_p x\| &\leq \|U \Pi_p x\| + \|\Pi_p U \Pi_p x - U \Pi_p x\| \\ &\leq \|U\|_{\text{op}} \|\Pi_p x\| + \Delta_p(U) \|\Pi_p x\| \\ &= (B + \Delta_p(U)) \|\Pi_p x\|, \end{aligned}$$

as claimed. □

C.2 Implication for Diagnostic $\Delta_{p,t}$

In the runtime scheme, we monitor

$$\Delta_{p,t} := \|U_t \Pi_p - \Pi_p U_t\|_{\text{op}}, \quad \Delta_t := \max_{p \in P} \Delta_{p,t},$$

for the audited update operator U_t at iteration t .

If the acceptance gate enforces $\Delta_t \leq \eta$ for some small $\eta > 0$, then for any state T_t with $\|T_t\| \leq R$ and any sector p , we have

$$\|\Pi_p U_t T_t - U_t \Pi_p T_t\| \leq \eta \|T_t\| \leq \eta R.$$

Thus the error in treating U_t as block-diagonal with respect to the sector decomposition is explicitly bounded; this underlies the empirical interpretation of Δ_t as a measure of “on-manifold” evolution and supports the use of Δ_t as a diagnostic for emergent modularity.

D Sector Shares, Manifold Energy, and Entropy

We formalize the definitions and bounds for the diagnostics $\text{ME}(t)$ and $H[q_t]$ used in the main text.

D.1 Sector Shares

Given a state $T \in \mathcal{H}$, we define the prime sector energies and shares.

Definition D.1 (Sector Energies and Shares). For $T \in \mathcal{H}$, define

$$E_p(T) := \|\Pi_p T\|^2, \quad E_{\text{tot}}(T) := \sum_{p \in P} E_p(T).$$

If $E_{\text{tot}}(T) > 0$, define the sector share vector $q(T) = (q_p(T))_{p \in P}$ by

$$q_p(T) := \frac{E_p(T)}{E_{\text{tot}}(T)}, \quad \sum_{p \in P} q_p(T) = 1.$$

If $E_{\text{tot}}(T) = 0$, set $q_p(T) = 0$ for all p .

D.2 Manifold Energy

Definition D.2 (Manifold Energy). Given $T \in \mathcal{H}$, we define the manifold energy (with a small regularization $\varepsilon > 0$) by

$$\text{ME}(T) := \frac{\|\Pi_P T\|^2}{\|T\|^2 + \varepsilon},$$

where $\Pi_P = \sum_{p \in P} \Pi_p$.

Proposition D.3 (Bounds on $\text{ME}(T)$). *For any $T \in \mathcal{H}$,*

$$0 \leq \text{ME}(T) \leq 1.$$

Moreover, if $\Pi_P T = T$, then

$$\text{ME}(T) = \frac{\|T\|^2}{\|T\|^2 + \varepsilon} \rightarrow 1$$

as $\varepsilon \rightarrow 0$.

Proof. Since Π_P is a projector with $\|\Pi_P\|_{\text{op}} \leq 1$, we have $\|\Pi_P T\| \leq \|T\|$. Thus

$$0 \leq \|\Pi_P T\|^2 \leq \|T\|^2,$$

and hence

$$0 \leq \frac{\|\Pi_P T\|^2}{\|T\|^2 + \varepsilon} \leq 1.$$

If $\Pi_P T = T$, then the numerator equals $\|T\|^2$, and the claimed limit is immediate. $\square$

D.3 Sector Entropy

Definition D.4 (Sector Entropy). Let $q = (q_p)_{p \in P}$ be a probability vector (sector shares). Define the Shannon entropy

$$H[q] := - \sum_{p \in P} q_p \log q_p,$$

using the convention $0 \log 0 := 0$.

Proposition D.5 (Entropy Bounds). *For any probability vector q supported on P ,*

$$0 \leq H[q] \leq \log |P|.$$

Moreover, $H[q] = 0$ iff q is a Dirac mass concentrated on a single prime sector, and $H[q] = \log |P|$ iff q is uniform:

$$q_p = \frac{1}{|P|} \quad \forall p \in P.$$

Proof. This is the standard Shannon entropy bound. The minimum at 0 occurs when one coordinate is 1 and the rest are 0, and the maximum at $\log |P|$ occurs at the uniform distribution. $\square$

In the runtime context, convergence of $H[q_t]$ to some finite value H_∞ indicates stabilization of the sector energy distribution; combined with $\text{ME}(T_t) \rightarrow 1$ and $\Delta_t \rightarrow 0$, this yields the diagnostic triad discussed in the main text.

E Hecke Operators on Truncated q -Series

We now give explicit operator norm estimates and algebraic identities for Hecke operators on truncated q -series.

E.1 Hecke Operator Definition

Fix a truncation parameter $N \in \mathbb{N}$. Consider the vector space $\mathcal{V}_N := \mathbb{C}^{N+1}$ with standard basis $e_0, \dots, e_N$ corresponding to coefficients $a(0), \dots, a(N)$ of a q -series.

Definition E.1 (Prime Hecke Operator on $\mathcal{V}_N$). For a prime p and a coefficient vector $a = (a(0), \dots, a(N)) \in \mathcal{V}_N$, define the truncated Hecke transform $T_p a \in \mathcal{V}_N$ by

$$(T_p a)[n] := \begin{cases} a[pn] + a[n/p] & \text{if } p \mid n, \quad pn \leq N, \\ a[pn] & \text{if } p \nmid n, \quad pn \leq N, \\ a[n/p] & \text{if } p \mid n, \quad pn > N, \\ 0 & \text{if } pn > N, \quad p \nmid n/p \text{ invalid.} \end{cases}$$

Equivalently, T_p is represented by an integer matrix $(T_p)_{n,m}$ with entries

$$(T_p)_{n,m} = \begin{cases} 1 & \text{if } m = pn \leq N, \\ 1 & \text{if } p \mid n \text{ and } m = n/p, \\ 0 & \text{otherwise.} \end{cases}$$

E.2 Operator Norm Bound (Euclidean Norm)

We equip $\mathcal{V}_N$ with the Euclidean norm $\|\cdot\|_2$. The matrix entries of T_p are in $\{0, 1\}$, and each row has at most two nonzero entries.

Proposition E.2 (Operator Norm Bound for T_p on $\mathcal{V}_N$). *Let T_p be defined as above, acting on $(\mathcal{V}_N, \|\cdot\|_2)$. Then*

$$\|T_p\|_{\text{op},2} \leq \sqrt{2}.$$

Proof. For each n , the n -th row of T_p has at most two entries in $\{0, 1\}$. Thus the squared Euclidean norm of each row is at most 2. The operator norm $\|T_p\|_{\text{op},2}$ is bounded by the maximum singular value, which in turn is bounded by the maximum row norm in Euclidean norm. Formally, if A is a matrix and r_i denotes the i -th row considered as a vector, then

$$\|A\|_{\text{op},2} \leq \max_i \|r_i\|_2.$$

Applying this to $A = T_p$ yields

$$\|T_p\|_{\text{op},2} \leq \max_n \sqrt{\#\text{nonzero entries in row } n} \leq \sqrt{2}.$$

□

For composite m , one can define T_m via convolution relations or by composing prime operators. If $m = \prod_{j=1}^r p_j^{e_j}$, the naive bound

$$\|T_m\|_{\text{op},2} \leq \prod_{j=1}^r \|T_{p_j}\|_{\text{op},2}^{e_j} \leq 2^{\frac{1}{2} \sum_j e_j}$$

is sufficient for many purposes; sharper estimates can be obtained by inspecting the combinatorial structure of the matrix.

E.3 Exact Commutativity and Multiplicativity on $\mathcal{V}_N$

In the infinite-dimensional setting of classical modular forms, Hecke operators satisfy exact commutativity and multiplicativity relations on the appropriate spaces of forms. On the truncated space $\mathcal{V}_N$, these identities hold exactly as long as the indices involved remain below the truncation.

Let T_m and T_n be constructed from the prime T_p matrices using the usual Hecke algebra relations.

Proposition E.3 (Exact Algebraic Relations Below Truncation). *Fix $N \in \mathbb{N}$. Let m, n be positive integers such that all indices arising in the action of the Hecke operators T_m, T_n, T_{mn} on coefficients $a(0), \dots, a(N)$ remain $\leq N$. Then on $\mathcal{V}_N$,*

$$T_m T_n = T_n T_m, \quad T_m T_n = T_{mn}.$$

Sketch of Proof. The proof proceeds by unraveling the combinatorial definition of the Hecke operators in terms of their action on coefficients and checking that, for indices below the truncation threshold, the compositions of the stencils match the usual Dirichlet convolution formulas. This is standard in the theory of Hecke operators on spaces of finite-support sequences. The key point is that no term “falls off” beyond N ; under that condition the truncated matrices reproduce the algebraic identities exactly. □

In practical computations, we verify these relations numerically (and, when using exact integer matrices, symbolically) inside a prescribed algebra window $\{m, n \leq M\}$ and a truncation N large enough that boundary effects are negligible for the indices under test.

F Summary of Bounds and Conditions

For ease of reference, we collect here the principal inequalities and conditions used in the development of the Multiplicity Moonshine validation engine:

- **Channel operator bound:** $\|K_t\|_{\text{op}} \leq \sum_{p \in P} b_p |w_{p,t}| \leq \tau_t < 1$.
- **Projected iteration contraction:** Under $\|\Pi\|_{\text{op}} \leq 1$, L_F -Lipschitz F_t , and ACE budget,
$$\|\Phi_t(T_1) - \Phi_t(T_2)\| \leq (L_F + \tau_t) \|T_1 - T_2\|.$$
- **Fixed point uniqueness and convergence:** If $L_F + \tau < 1$ (uniform in t), then the iteration admits a unique fixed point T^* , and
$$\|T_t - T^*\| \leq (L_F + \tau)^t \|T_0 - T^*\|.$$
- **Commutation residual:** $\|\Pi_p Ux - U\Pi_p x\| \leq \Delta_p(U) \|x\|$, with $\Delta_p(U) = \|U\Pi_p - \Pi_p U\|_{\text{op}}$.
- **Manifold energy bound:** $0 \leq \text{ME}(T) = \frac{\|\Pi_P T\|^2}{\|T\|^2 + \varepsilon} \leq 1$.
- **Entropy bound:** $0 \leq H[q(T)] \leq \log |P|$.
- **Hecke prime operator norm:** $\|T_p\|_{\text{op},2} \leq \sqrt{2}$ for the truncated stencil described above.
- **Hecke algebra relations (within truncation window):** $T_m T_n = T_n T_m$, $T_m T_n = T_{mn}$ on $\mathcal{V}_N$ as long as the indices involved do not exceed N .

These bounds and identities formalize the analytic backbone of the Multiplicity Moonshine validation protocol: they guarantee that the iterated map is well-behaved (contraction), that sectorwise structure is stable (small commutation residual), that energy and entropy diagnostics are meaningful and bounded, and that the discrete Hecke algebra employed in computations mirrors the continuous algebraic structure of modular forms within a controlled truncation regime.

References

- [Atkin and Lehner(1970)] A. O. L. Atkin and Joseph Lehner. Hecke operators on $\gamma_0(m)$. *Mathematische Annalen*, 185(2):134–160, 1970. doi: 10.1007/BF01359701.
- [Banach(1922)] Stefan Banach. Sur les operations dans les ensembles abstraits et leur application aux equations integrales. *Fundamenta Mathematicae*, 3:133–181, 1922.
- [Borcherds(1992)] Richard E. Borcherds. Monstrous moonshine and monstrous lie superalgebras. *Inventiones Mathematicae*, 109(2):405–444, 1992. doi: 10.1007/BF01232032.
- [Borcherds(1998)] Richard E. Borcherds. Automorphic forms with singularities on grassmannians. *Inventiones Mathematicae*, 132(3):491–562, 1998. doi: 10.1007/s002220050232.
- [Cheng et al.(2014)Cheng, Duncan, and Harvey] Miranda C. N. Cheng, John F. R. Duncan, and Jeffrey A. Harvey. Umbral moonshine. *Communications in Number Theory and Physics*, 8(2):101–242, 2014. doi: 10.4310/CNTP.2014.v8.n2.a1.

- [Conway and Norton(1979)] John H. Conway and Simon P. Norton. Monstrous moonshine. *Bulletin of the London Mathematical Society*, 11(3):308–339, 1979. doi: 10.1112/blms/11.3.308.
- [Diamond and Shurman(2005)] Fred Diamond and Jerry Shurman. *A First Course in Modular Forms*, volume 228 of *Graduate Texts in Mathematics*. Springer, 2005. doi: 10.1007/b138964.
- [Duncan and Frenkel(2011)] John F. R. Duncan and Igor B. Frenkel. Rademacher sums, moonshine and gravity. *Communications in Number Theory and Physics*, 5(4):849–976, 2011. doi: 10.4310/CNTP.2011.v5.n4.a4.
- [Frenkel et al.(1988)Frenkel, Lepowsky, and Meurman] Igor B. Frenkel, James Lepowsky, and Arne Meurman. Vertex operator algebras and the Monster. *Pure and Applied Mathematics*, 134, 1988.
- [Gannon(2006)] Terry Gannon. *Moonshine Beyond the Monster: The Bridge Connecting Algebra, Modular Forms and Physics*. Cambridge Monographs on Mathematical Physics. Cambridge University Press, 2006. doi: 10.1017/CBO9780511617083.
- [Hecke(1937)] Erich Hecke. Neue Theorie der elliptischen Modulfunktionen von zwei Variablen und ihrer Darstellung durch eine neue Art von Dirichletschen Reihen. *Mathematische Annalen*, 114:1–28, 1937.
- [Iwaniec(2002)] Henryk Iwaniec. *Spectral Methods of Automorphic Forms*, volume 53 of *Graduate Studies in Mathematics*. American Mathematical Society, 2002.
- [Johansson(2017)] Fredrik Johansson. Arb: Efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Transactions on Computers*, 66(8):1281–1292, 2017. doi: 10.1109/TC.2017.2690633.
- [Johansson and contributors()] Fredrik Johansson and contributors. python-flint: Python bindings for flint and arb. <https://pypi.org/project/python-flint/>. Accessed 2026.
- [Johansson et al.()] Fredrik Johansson et al. mpmath: A Python library for arbitrary-precision floating-point arithmetic. <https://mpmath.org/>. Accessed 2026.
- [LMFDB Collaboration()] LMFDB Collaboration. The L-functions and modular forms database. <https://www.lmfdb.org/>. Online database of L -functions, modular forms, and related objects.
- [Meurer et al.(2017)] Aaron Meurer et al. *SymPy: Symbolic Computing in Python*. PeerJ Computer Science, 2017. Software paper; SymPy project.
- [Miyake(2006)] Toshitsune Miyake. *Modular Forms*. Springer Monographs in Mathematics. Springer, 2006. doi: 10.1007/3-540-29593-4.
- [Multiplicity Foundation(2026a)] Multiplicity Foundation. Az-tftc and tensorial multiplicity. Internal preprint, Citizen Gardens / Multiplicity Foundation, 2026a. Tensorial time-frequency constructions and multiplicity invariants.
- [Multiplicity Foundation(2026b)] Multiplicity Foundation. Ftsa, zsd, and prime-indexed spectral architectures. Internal preprint, Citizen Gardens / Multiplicity Foundation, 2026b. Frequency-temporal spectral architectures and zero-shot dynamics in prime-indexed spaces.
- [Multiplicity Foundation(2026c)] Multiplicity Foundation. The langlands multiplicity prism. Internal preprint, Citizen Gardens / Multiplicity Foundation, 2026c. Prime-word grading, multiplicity transfer, and Langlands-style correspondences.

- [Multiplicity Foundation(2026d)] Multiplicity Foundation. Meta-relativity and g-theory. Internal preprint, Citizen Gardens / Multiplicity Foundation, 2026d. Lawfulness projectors, recursive tensor structures, and multiplicity in generalized relativity.
- [Multiplicity Foundation(2026e)] Multiplicity Foundation. Multiplicity core for moonshine — mathematical blueprint. Internal preprint, Citizen Gardens / Multiplicity Foundation, 2026e. Prime projectors, ACE budget, PETC ledger, certified fixed-point engine for Moonshine workloads.
- [Multiplicity Foundation(2026f)] Multiplicity Foundation. Neuromorphic ume and multiplicity dynamics. Internal preprint, Citizen Gardens / Multiplicity Foundation, 2026f. Neuromorphic universality, multiplicity operators, and runtime-safe evolution.
- [Multiplicity Foundation(2026g)] Multiplicity Foundation. Phasemirror-hq. <https://github.com/MultiplicityFoundation/PhaseMirror-HQ>, 2026g. Codebase implementing the multiplicity core, PETC, ACE, and Moonshine workloads.
- [Multiplicity Foundation(2026h)] Multiplicity Foundation. Prime-colored braid multiplicity. Internal preprint, Citizen Gardens / Multiplicity Foundation, 2026h. Prime-colored braids, Yang–Baxter structures, and multiplicity invariants.
- [Norton(2001)] Simon P. Norton. Generalized moonshine. In Arne Meurman et al., editors, *The Monster and Lie Algebras*. Walter de Gruyter, 2001. Preprint circulated earlier; standard reference for generalized Moonshine.